

Embedded Keccak implementation on FPGA

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Abstract— The use of Blockchain has evolved over the last decade. It has touched many fields such as health, industry 4.0, smart cities, agriculture, education etc. Being a guarantee of security, the Blockchain requires to be careful with the data used. In the medical field, patient data is confidential and in order to have access to it, it is necessary to guarantee foolproof security. To do this, the best consensus to use is the PoW. In this paper, we will optimize the code and implement the PoW as an IP in order to accelerate its execution time and consequently minimize the energy consumption. The used Blockchain in this work is the Ethereum and its consensus is the Keccak.

Keywords—Blockchain, PoW, Ethereum, Keccak, Embedded System, FPGA

I. INTRODUCTION

With the evolution of telecommunication and particularly the generation of mobile networks (3G, 4G etc.), the IoT has evolved and has been used in several application areas. Thus we have seen the IoT [24] used in the industrial field and particularly in Industry 4.0 [1-2] where specific sensors have appeared. In this case, we speak of IIoT [3].

In the medical field, we speak of IoMT [4,19]. While the backup of the various data can be done using the cloud, the security [21-22] of these resources is a problem in certain very critical applications [5,17]. To secure this information and to allow the tracking of the different data, we use the Blockchain technology. In this work we will present this technology and particularly the Ethereum Blockchain [6].

Ethereum is a permissioned Blockchain. It is based on Proof of Work (PoW) consensus. This consensus is very greedy in terms of energy consumption and execution time. PoW is used during the mining process.

To secure the Blockchain, we use the consensus. Blockchain permits to secure the data. That's why we use a public-private key system to encrypt transactions. These transactions are gathered into a bloc. To create a new block, we use the consensus such as Proof of Work. The PoW is based onto Keccak Algorithm [7,14]. Being very greedy, we will implement an optimized HW implementation of the Keccak consensus to improve the consumption and speed the proposed system [7,12]. The HW implementation is proposed using VHDL.

After optimizing the Keccak Software, we implement the hardware IP on FPGA and particularly on FPGA. The used platform is the Xilinx ML 507. [8,18,20] We chose the ML 507 platform, which is still a fairly old platform for two

reasons. Indeed, this platform offers processors with a frequency that does not exceed 150 Mhz, which allows us to highlight the quality of the architecture that has been realized in this paper.

On the other hand, this platform is the only one which is available to make a real implementation of the consensus in FPGA.

This paper presents the following plan :

- ✓ We start with the state of the state of the art chapter which proposes the different uses of the Blockchain.
- ✓ In the next part, we will describe the Keccak consensus used and implemented in HW.
- ✓ In part 3 we describe and analyze the obtained results after FPGA implementation.
- ✓ We finish this article with a conclusion and some perspectives.

II. EASE OF USE

In this section, we will describe a state of the art. This state of the art is in two parts. First, we will present the Blockchain technology. Then, we will highlight the consensus algorithms.

A. Blockchain

Blockchain technology is based on the same principle as Peer to Peer (P2P) networks [9]. Indeed, each actor trusts the blockchain composed by all the actors even if they do not trust each other. We can therefore describe the blockchain as a shared and reliable archive of operations (transactions, data, etc.). The most important specificity of this archiving is that when a copy is created, it becomes immutable. This information is encrypted and stored on each node of the network to be used. The Blockchain can be compared to a marble on which one engraves information. The range obtained engraving is irreversible. One can modify the information on the marble by adding other data but one does not have the possibility of removing what was previously engraved. Thus, it is impossible to corrupt a data that has been registered on our Blockchain.

Blockchain is based on a specific terminology representing important concepts. [6] Among these terms we can mention:

- Transactions: They constitute the data exchanges made between different users. Each transaction is signed by

the private key of the sender. Thanks to this signature, the security of the transactions is guaranteed. Consequently, it is possible to avoid any modification of these transactions during the transmission. [9-10]

- Blocks: A block is a record in the Blockchain that contains confirmed transactions. Thus, each current transaction will be added to a block. After a period, for a new block containing transactions to be added to the Blockchain, it must be validated by a selected person called a miner. This validation operation is called mining. The block is validated by using a consensus. [9-10]

- Chained Blocks: Each block of the Blockchain is linked to the previous block. This link is done by inserting the hash specific to the previous block. Therefore, the hash of each block includes not only the hash of the block itself but also the hash of the previous block. This way we can protect our blockchain from any form of corruption. This prevents any modification of the content of existing blocks. [9-10]

- Smart contract : A smart contract is a software "installed" on a blockchain solution. It is the most important link in our blockchain. It runs automatically as soon as the various pre-programmed constraints are verified. Although it is not a legal document in itself, the smart contract automates the execution of a contractual commitment. [9-10]

B. Consensus Algorithm

A secure chain is a single master chain whose state is consistent. Each valid block added to this chain adds to the overall trust level of the chain. Miners compete to add their block to the chain. What happens if everyone wants to add their candidate block to the chain?

Each candidate block is none other than that of a competing miner. In this context, several questions arise: What is the next block to be added to the chain? Can our miners agree on the next block? Is there a method or a protocol to choose the next block?

The answer to these questions is yes. There is, in fact, a protocol that allows this task to be done. It is called Proof of Work (PoW). PoW [5,12] uses a method that was discussed earlier. It is the versatile hash technology.

Let's now talk about Proof of Work as it is used in Bitcoin and Ethereum. Let's introduce this approach from the miner's perspective. First of all, we need to calculate the hash of the block header elements which is a fixed value, and a nonce which is a variable. If the hash value is less than 2 by 128 for Bitcoins, and less than the difficulty function for Ethereum blockchains, the puzzle will be solved. Otherwise, the process must be repeated after changing the nonce value.

If the puzzle has been solved, the winning block should be broadcast and verified by the other miners. The non-winning miner nodes add the new block to the local copy of the chain and move on to work on the next block. The winner receives a reward for creating the block. Proof of work is a consensus protocol used by the Bitcoin and Ethereum blockchain.

Several protocols exist today, but the most used are proof of work (PoW), proof of stake (Pos), and proof of authority (PoA) which is a more recently created protocol than both PoW and PoS.

In Table 1, we compare the most known consensus.

Protocol	Advantages	Disadvantages
PoW	- Safe - Stable - High degree of freedom of nodes - High degree of decentralization, open node system	- Weak scalability - low performance - Causing hardware equipment waste - High energy consumption
PoS	- Less energy - High degree of decentralization, open node system	- Complex implementation process - Security breach
PoA	- Less energy - High performance - Finality	- Weak degree of decentralization - closed node system

As shown in the table, we notice that PoW remains the most greedy, most powerful and most secure consensus. For this reason we find that this heavy consensus is still used by Bitcoin, Ethereum and other public blockchains to maintain the security of the information.

The proof of stake commonly called the consensus of the rich is generally used in private blockchains. This consensus allows the richest nodes (which have the most computing power) to validate block creation. This validation is therefore done thanks to the power and not via the resolution of the PoW.

PoA is more secure than PoS but less efficient than PoW. This being said, it has a very limited consumption compared to PoW. Hence the usefulness of using it especially in the framework of authorized Blockchains.

III. CONSENSUS IMPLEMENTATION

Keccak is a hash function based on the sponge construction. It is considered as a sponge function family. Keccak has several parameters that can be chosen by the user.

The underlying function is a permutation chosen in 7 Keccak- f permutations, denoted Keccak- $f[b]$.

$b \in \{25, 50, 100, 200, 400, 800, 1600\}$ represents the width of the permutation.

Organization of the state as an array of 5×5 lanes, each of length $w \in \{1, 2, 4, 8, 16, 32, 64\}$ and $b = 25w$. When implemented on a 64-bit processor, a lane of Keccak- $f[1600]$ can be represented as a 64-bit CPU word. [11,13]

In figure 2, we represent the keccak different functionalities.

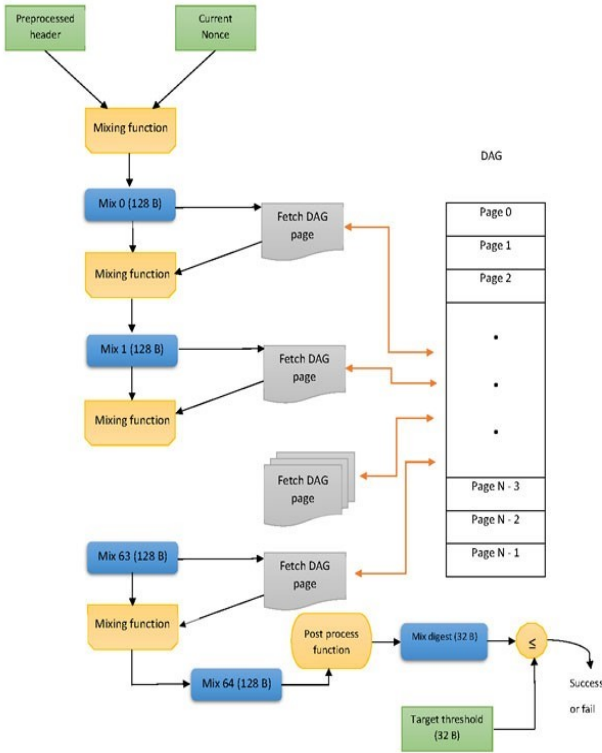


Fig. 1. Flowchart of the Keccak 256 algorithm used by Ethereum.

IV. OBTAINED RESULTS

A. SW implementation

When computing the hash algorithm, the majority of time is spent on Keccak-f. Thus we choose it to be our accelerator.

If Keccak is used as SHA-3, the state is 1600 bits which is stored in 25 words of 64 bits each. On 64 bit CPUs, which are in the majority of modern PCs, one 64 bit lane can be stored naturally in one register.

Generally speaking, Keccak is quite amenable to software implementation. It shares this property with the other SHA hash algorithms. A highly optimized SHA-3

implementation on modern Intel Core CPUs can be executed at a rate of about 13 cycles/byte which translates, e.g., to a throughput of approximately 230 MByte/s (or about 1.84 Gbit/s) if the processor is clocked at 3 GHz.

On 8-bit CPUs, which are very popular in embedded systems, SHA-3 can be implemented at about 1110 cycles/byte. Assuming a clock frequency of 10 MHz, this results in a throughput of about 9 kByte/s, or roughly 72 kbit/s.

For our case, we run our implementation on a 7th Gen Intel Core i7 CPU over clock 3.5 GHz..

In Table 2, we represent the Software implementation on both CPU using native C++ and GPU platform.

The GPU implementation time is smaller than the CPU one. It is due to the parallelism obtained via GPU. In this work we use the PC GPU. This GPU has a 1.665 Ghz frequency.

TABLE II. SW IMPLEMENTATION

	Software	
	CPU (C++)	GPU
Execution time (ms)	0.021	0.0086

In part B, we will implement a HW architecture. This implementation will be simulated using ModelSim SW then we will implement this IP on Virtex 5 ML 507 FPGA platform.

B. HW implementation

In this part, we have transformed the code of our implemented PoW into HW architecture.

This architecture has been implemented in VHDL. Then we simulated the implemented code using the simulation tool ModelSim. In figure 1, we propose a screenshot of simulation.

This figure 2 shows :

- the input signal as the third signal (clearer in the zoomed in figure) .
- The output signal is the last shown signal (red color) .

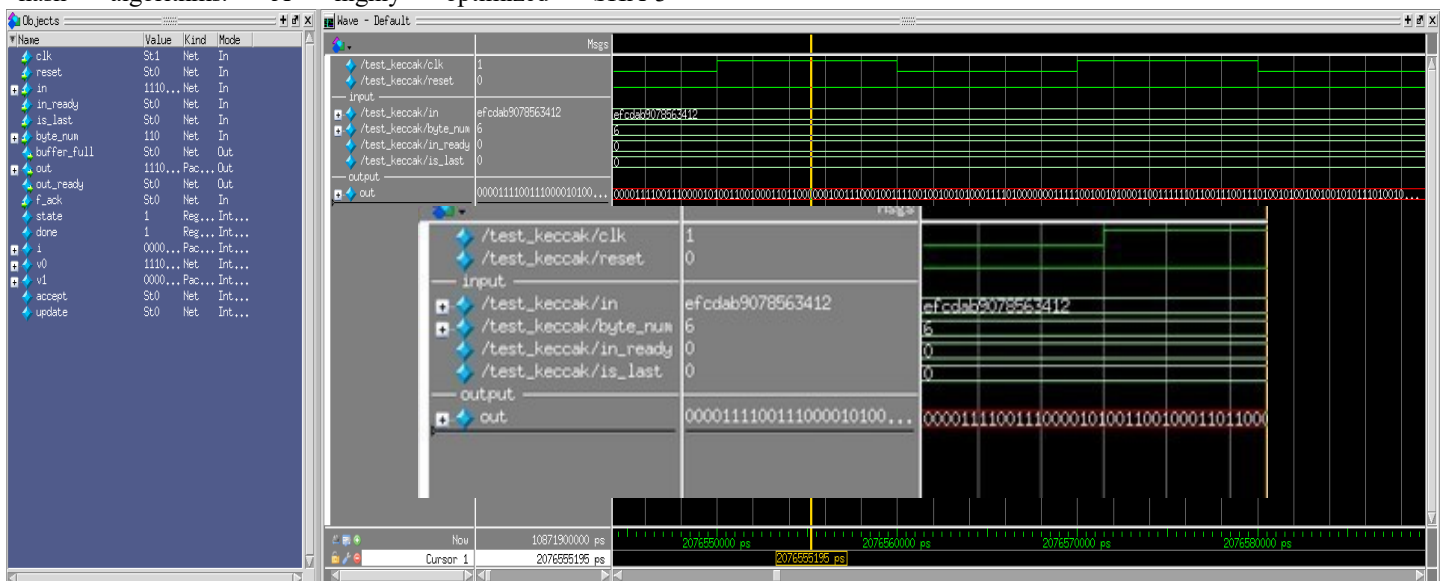


Fig.2 Keccak HW simulation

following sentence as input “ The quick brown fox jumps over the lazy dog “for hash.

The result of this simulation is added to Table 1 in order to show the importance of the implementation of our optimized IP. Table 3 illustrates a comparison between SW& HW implementation.

Despite the low frequency of our FPGA, we notice that the HW execution time is approximately 1/3 of the GPU execution time.

The code implemented on our GPU was parallelized but the result obtained on FPGA allowed having a faster execution time. This execution time is due to the HW optimization.

TABLE III. COMPARISON BETWEEN SW & HW IMPLEMENTATION

	Hardware	Software	
	<i>FPGA implementation</i>	<i>CPU (C++)</i>	<i>GPU</i>
Execution time (ms)	0.00398	0.021	0.0086

II. CONCLUSION & PERSPECTIVES

The emergence of blockchain has made it one of the references in terms of data security, information traceability and transaction backup. For the information protection, the PoW is the best consensus to put in place. Having presented the PoW encryption algorithm based on Keccak, we described in this paper the implementation of a PoW-based IP implemented on FPGA. The optimized IP allows having a lower execution time than the classical SW implemented on an i7 processor and on a GPU.

Having demonstrated the importance of using a VHDL architecture, the best perspective to this work would be to implement a multi IP system allowing to parallelize the PoW implementation. The use of Dynamic Partial Reconfiguration [15,20] can save energy depending on the complexity of the consensus. It is also possible to use approximate computing to further minimize energy and resource consumption [16,23]

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